

TYLER DOESE

Mechanical Engineer • Design & FEA
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[Portfolio](#) • [YouTube](#) • [LinkedIn](#)

SUMMARY

Mechanical engineering graduate pairing hands-on automotive service experience with applied SolidWorks CAD and FEA design work. Seeking an entry-level role in the performance aftermarket, where powertrain knowledge, fabrication ability, and design analysis intersect.

EDUCATION

Kennesaw State University — B.S., Mechanical Engineering May 2026

GPA: 3.5

- **Relevant coursework:** Machine Design, Finite Element Analysis, Heat Transfer & Thermodynamics, Internal Combustion / Fluid Mechanics, Manufacturing Processes, Controls.

PROJECTS

Variable-Width Automotive Wheel — Senior Capstone Design

- Designed a pneumatically actuated, variable-width wheel using twin shells (fixed inner, adjustable outer) linked by five telescoping cylinders and a central actuator.
- Built the full assembly in SolidWorks and ran FEA across static radial, cornering, and centrifugal load cases on 7075-T6 aluminum, validating stress and displacement margins.

Pumpkin Trebuchet — KSU Pumpkin Launch (1st Place)

- Led structural and throwing-mechanism design and lead-presented proposal/design reviews; managed team finances and material procurement. Winning launch: 390 ft.

High-Precision Ping-Pong Launcher — Machine Design Competition (2nd Place)

- Performed calculations and analysis for primary machine components; designed and 3d printed several parts.

ENGINEERING EXPERIENCE

Cleveland Electric May 2025 – Jul 2025

Project Management Intern

- Performed on-site floor walks to monitor project progress, verify installed work, and surface field issues for the project management team.
- Supported contractor coordination meetings, tracking scope, schedule, and open action items across trades.
- Estimated cost and labor impacts of work-order changes to inform change-order pricing and budget decisions.

RoviSys May 2023 – Aug 2023

Engineering Intern — Manufacturing Systems & Controls Integration

- Redesigned electrical control cabinets in SolidWorks for manufacturing automation systems.
- Supported controls integration and designed Human-Machine Interface (HMI) layouts for production equipment.

AUTOMOTIVE EXPERIENCE

Express Oil Change & Tire Engineers Woodstock, GA

Mechanic Jun 2025 – Present

- Diagnose and repair engine, brake, suspension, and drivetrain faults using scan tools and systematic troubleshooting — beyond routine maintenance.
- Perform full mechanical repairs and component replacement, then verify each fix through road testing and re-inspection.

Automotive Technician Dec 2023 – Oct 2024

- Performed scheduled maintenance, diagnostics, and component replacement across a high volume of customer vehicles.
- Generated repair quotes and communicated technical findings directly to customers in clear, actionable terms.

SKILLS & TOOLS

- **CAD & Analysis:** SolidWorks (modeling, assemblies, FEA / Simulation), GD&T, hand calculations
- **Automotive & Powertrain:** Engine assembly & diagnostics, drivetrain design
- **Additive Manufacturing:** FDM 3D printing (PLA/PETG), CAD, slicing
- **Programming:** MATLAB, Simulink, VBA, Java; ladder logic / linear controls